

Intelligence as Representation-Solver Compatibility

A General Theory of Representation-Dependent Reasoning

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Abstract

No existing framework formally explains *why* the same AI system succeeds on one encoding of a problem and fails on another—or predicts, before execution, which failures are recoverable and which are not. We address this gap by proving that problem difficulty is not intrinsic to tasks but *relational*: a function $D(P, E, S)$ of the problem P , encoding E , and solver S jointly.

We present **Representation-Solver Compatibility Theory (RSCT)**, which certifies whether an encoding is admissible for a given solver before execution. RSCT decomposes representation quality into three structurally independent axes: (i) signal purity $\alpha = R/(R + N)$, grounded in Fano’s inequality; (ii) geometric compatibility κ_{gate} , measuring the fraction of solver potential the encoding unlocks; and (iii) representational turbulence σ , governing dynamical stability.

The central result—Theorem 4.1—is that intelligence, precisely defined, *is* encoding competence: the fraction of a solver’s theoretical ceiling that a given encoding policy achieves. Capacity (the solver’s ceiling) is a fixed parameter of the deployment context, not a controllable variable. When competence is low, scaling capacity produces bounded, predictable diminishing returns (Corollary 4.1.1). This is a falsifiable, quantitative prediction about the limits of the prevailing scaling paradigm.

From this foundation we derive: (i) three structural independence theorems proving that no two of $(\alpha, \kappa_{\text{gate}}, \sigma)$ constrain the third; (ii) a weakest-link optimality theorem proving that min-aggregation is the *unique* adversarially safe composition rule for modal health scores; (iii) a four-gate pre-execution gatekeeper with $O(M)$ complexity that enforces all three axes sequentially; and (iv) a theoretical reframing of learning dynamics as Representation Search under Stability Constraints (RSSC), showing how training can be viewed as search over encoding space. The operational DGM implementation uses fixed embeddings and geometric compatibility without runtime learning.

Keywords: Representation Learning, Compatibility Theory, Hallucination, AI Certification, Fano Bound, Scaling Laws, Curriculum Learning, Model Risk Management, SR 11-7, Representation Search.

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1 The Intrinsic Difficulty Assumption and Why It Fails

Difficulty is a relationship between a problem, an encoding, and a solver — not a property of the problem alone.

1.1 The Dominant Paradigm’s Hidden Assumption

Every major AI benchmark treats problem difficulty as intrinsic to the task. GSM8K problems are “harder” than single-digit arithmetic. MMLU is “harder” than true-or-false tests. Models that solve more “harder” problems under fixed compute budgets are deemed more intelligent. Underperformance is attributed to insufficient capacity: more parameters, more compute, larger context windows.

This framing rests on an assumption that is almost never stated explicitly: that $D(P) = D(P, E, S)$ — that difficulty is a property of the problem alone, independent of how it is encoded and which solver processes that encoding. RSCT proves this assumption is false, and quantifies the consequences of holding it.

1.2 Evidence of Encoding Dominance

Three domains provide decisive evidence that encoding choice dominates solver choice.

Sudoku. The underlying problem P is a unique completed 9×9 grid. The encoding E is the mask of revealed givens. Mathematical proof [1] shows that thousands of essentially different 17-clue masks exist for the same grid; which mask is chosen determines whether the instance is trivially solvable by cascade deduction or computationally intractable. The solver changes nothing.

Max-Cut and VQE. A Hamiltonian that traps a Variational Quantum Eigensolver in barren plateaus under one ansatz circuit encoding converges rapidly under an alternative circuit topology for the same problem [2]. The physics does not change; the encoding does.

Graph connectivity. An adjacency list and a prose description of the same graph carry identical information ($R = 0.9, N = 0.1$ under either encoding). For a graph-traversal solver, the adjacency list achieves $\kappa \approx 1$; the prose encoding forces the solver to parse structure from natural language, yielding $\kappa \ll 1$. Identical purity; wildly different compatibility.

Observation 1.1 (Encoding Dominance). *Across multiple domains, representation choice produces 10–1000× performance variation. Problem-intrinsic features and solver choice together produce 2–5× variation under identical compute budgets.*

1.3 The Central Claim

If difficulty is relational rather than intrinsic, intelligence cannot be defined by solver capacity alone. It must encompass *encoding competence*: the ability to construct or discover representations that achieve high compatibility with the solver. RSCT formalizes this precisely.

The paper proceeds as follows. Section 2 establishes the YRSN signal purity foundation (the α axis). Section 3 establishes geometric compatibility (the κ_{gate} axis) and turbulence (the σ axis), and characterizes the admissible encoding region. Section 4 proves the four core theorems: the Capacity–Competence Factorization, the three structural independence results, and the weakest-link optimality result. Section 5 derives the four-gate pre-execution gatekeeper as the necessary enforcement consequence of those

theorems. Section 6 formalizes training as representation search (RSSC). Section 7 demonstrates substrate-independence across five domains.

2 Foundation I: Signal Purity (YRSN)

Distraction wastes context. Noise corrupts it.

When noise dominates, no solver regardless of capacity can recover correct inference.

2.1 The YRSN Decomposition

Any context C presented to a solver is a mixture of three structurally distinct components.

Definition 2.1 (Signal Purity Decomposition). *A context C decomposes as (R, S_{sup}, N) where:*

- R (**Relevant signal**): *tokens carrying mutual information with the latent problem structure X . Directly useful.*
- S_{sup} (**Superfluous content**): *tokens with negligible mutual information with X . Occupy context budget but do not corrupt.*
- N (**Adversarial noise**): *tokens anti-correlated with correct inference. Actively mislead; do not merely dilute.*

Subject to simplex normalization: $R + S_{\text{sup}} + N = 1$, with $R, S_{\text{sup}}, N \geq 0$. S_{sup} is never abbreviated to S (reserved throughout for Solver).

Definition 2.2 (Signal Purity).

$$\alpha = \frac{R}{R + N} \in [0, 1]. \quad (1)$$

Purity is defined over R and N only. S_{sup} does not appear because superfluous content does not corrupt the information channel between C and X ; it dilutes but does not deceive.

Remark 2.1 (Why $S_{\text{sup}} \geq 0.5$ Does Not Trigger a Hard Reject). *Gate 1 of the gatekeeper (Section 5) fires on $N \geq 0.5$, not on $S_{\text{sup}} \geq 0.5$. This asymmetry is not a design choice; it follows from the information theory. Superfluous content satisfies $I(X_S; X) \approx 0$: it occupies context but does not increase the probability of error. Noise satisfies $I(X_N; X) > 0$ with anti-correlation: it actively increases P_e . Fano's inequality applies to N , not to S_{sup} .*

2.2 Information-Theoretic Grounding

Let X denote latent problem structure and C the observed context, generated by a mixture of three independent channels: a relevant channel (X_R , high $I(X_R; X)$), a superfluous channel (X_S , $I(X_S; X) \approx 0$), and a noise channel (X_N , positive $I(X_N; X)$ but anti-correlated with correct inference).

Proposition 2.1 (Purity as Signal-to-Interference Ratio). *Under the channel mixture model, $\alpha = \text{SIR}(C, X)$ where*

$$\text{SIR}(C, X) := \frac{I(X_R; X)}{I(X_R; X) + I(X_N; X)}. \quad (2)$$

Proof. $\alpha = R/(R + N) = I(X_R; X)/(I(X_R; X) + I(X_N; X))$ by the proportionality assumption. Since $I(\cdot; \cdot) \geq 0$, we have $\alpha \in [0, 1]$. $\square$

Proposition 2.2 (Fano Bound on Solver Performance). *For any solver S operating on context C to recover latent structure $X \in \mathcal{X}$, Fano’s inequality gives:*

$$P_e \geq 1 - \frac{I(X; C) + 1}{\log |\mathcal{X}|}. \quad (3)$$

Under the channel mixture model, the exploitable mutual information satisfies $I_{\text{exploit}}(X; C) \leq \alpha \cdot H(X)$. This bound holds because the noise channel’s mutual information with X is anti-correlated with correct inference: it increases the solver’s probability of error rather than providing a recoverable signal. A solver that could invert the noise channel’s anti-correlation would, by definition, reclassify those tokens as relevant signal (raising α), contradicting the YRSN decomposition. Substituting:

$$P_e \geq 1 - \frac{\alpha \cdot H(X) + 1}{\log |\mathcal{X}|}. \quad (4)$$

As $\alpha \rightarrow 0$, $P_e \rightarrow 1 - 1/\log |\mathcal{X}|$ regardless of solver capacity.

This result is an *information-theoretic impossibility*, not a performance guideline. When α is near zero, the question “which solver should we deploy?” has no good answer — because the information required for correct inference is not present in the encoding. The right question is “how do we raise α ?” Gate 1 of the gatekeeper exists precisely because of this result.

2.3 Reliability-Weighted Quality and Temperature

When the context was drawn from a distribution that differs from the solver’s training distribution, an additional degradation factor applies.

Definition 2.3 (Reliability-Weighted Quality). *Let $\omega \in [0, 1]$ denote distributional reliability (OOD confidence, estimated from the solver’s latent representation). Define:*

$$\alpha_\omega = \omega\alpha + (1 - \omega)\alpha_{\text{base}}, \quad (5)$$

where $\alpha_{\text{base}} \in [0, 1]$ is a fixed robustness fallback (default $\alpha_{\text{base}} = 0.5$).

Definition 2.4 (Temperature). $\tau = 1/\alpha_\omega \in [1, \infty)$. *Temperature tracks task effective complexity: as encoding quality degrades, τ rises. Temperature governs curriculum pacing and quality-phase transitions (HIGH/MEDIUM/LOW). It is a scheduling instrument, not an enforcement construct; it does not appear in formal gates or theorems.*

3 Foundation II: Geometric Compatibility and Turbulence

Compatibility measures how much of the solver’s potential the encoding unlocks. A perfect solver on an incompatible encoding performs no better than a weak one.

3.1 Relational Difficulty and Atomic Compatibility

Definition 3.1 (Relational Difficulty). *The difficulty of problem P under encoding E for solver S is a triadic functional $D(P, E, S) \in (0, \infty)$. The optimal difficulty under the best possible encoding is $D^*(P, S) := \inf_E D(P, E, S)$.*

Definition 3.2 (Atomic Compatibility).

$$\kappa(E, S) := \frac{D^*(P, S)}{D(P, E, S)} \in [0, 1]. \quad (6)$$

$\kappa = 1$ iff the encoding achieves optimal difficulty for the solver. $\kappa \approx 0$ means the encoding forces the solver to work exponentially harder than necessary.

3.2 Tripartite Compatibility Architecture

Three levels of κ serve distinct purposes and must not be conflated.

Definition 3.3 (Modal Health and Execution Score). *Given a compositional encoding with M modal sources, each contributing a local health score $\kappa_i \in [0, 1]$:*

- κ_H : High-level health (text, structured data). Mean over high-level modalities.
- κ_L : Low-level health (vision, audio, video, sensor). Minimum over low-level modalities.
- κ_{int} : Interface compatibility. The minimum cross-modal alignment score from the φ -morphism mapping representation nodes to solver nodes.
- $\kappa_{\text{gate}} := \min(\kappa_H, \kappa_L, \kappa_{\text{int}})$: Enforced execution score. Used exclusively for real-time gating.

Remark 3.1 (Diagnostic Simplex — Post-Hoc Only). *A separate tuple $(\kappa_A, \kappa_E, \kappa_T) \in \Delta_2$ with $\kappa_A + \kappa_E + \kappa_T = 1$ decomposes the compatibility gap $1 - \kappa$ into Algorithmic, Encoding, and Tractability contributions. This diagnostic simplex is strictly post-hoc and is never used in real-time gating decisions.*

3.3 The Admissible Encoding Region

Definition 3.4 (Admissible Encoding Region). *The admissible region $\mathcal{E}_{\text{adm}}(S, P) \subset \mathcal{E}$ contains encodings for which solver dynamics $z_{t+1} = \mathcal{F}_S(z_t, E(P), \xi_t)$ remain bounded and convergent. Equivalently, these are encodings for which trajectories reach the solution attractor z^* rather than diverging.*

Definition 3.5 (Representational Turbulence).

$$\sigma(E, S) := \text{std}(\{\kappa_i\}_{i=1}^M) \in [0, 1]. \quad (7)$$

Turbulence measures how scattered the modal health scores are: low σ means all modalities contribute consistently; high σ means the encoding is geometrically incoherent across modalities. The stability requirement is $\sigma \leq \sigma_{\text{thr}}$ (default $\sigma_{\text{thr}} = 0.5$).

Lemma 3.1 (Turbulence Monotonicity). *Under mild Lipschitz regularity on $\mathcal{F}_S$, replacing a fraction $\epsilon > 0$ of relevant signal tokens with adversarial noise tokens increases turbulence: $\sigma(E_2, S) \geq \sigma(E_1, S)$.*

Proof sketch. Adversarial replacement targets the modality i^* whose tokens are most accessible to the perturbation, degrading κ_{i^*} while leaving the remaining κ_j ($j \neq i^*$) unchanged. Under Lipschitz continuity of $\mathcal{F}_S$ with constant L , the degradation satisfies $\kappa_{i^*}(E_1) - \kappa_{i^*}(E_2) \geq \epsilon L > 0$. Since $\sigma = \text{std}(\{\kappa_i\})$, reducing a single component while holding others fixed strictly increases the dispersion: the distance from κ_{i^*} to the mean increases while other distances remain constant. Hence $\sigma(E_2, S) > \sigma(E_1, S)$. $\square$

This lemma establishes that turbulence is not an independent noise source but a geometric signature of encoding degradation. It motivates Gate 3’s Oobleck principle (Section 5): as the encoding degrades, turbulence rises and the admissibility threshold rises with it.

3.4 The $\mathbb{T}^4$ Stability Representation

Namespace rule (mandatory). Certificate scalars $(\alpha, \omega, \alpha_\omega, \kappa_{\text{gate}}, \sigma)$ are real-valued quality measures in $[0, 1]$. $\mathbb{T}^4$ geometry uses angular coordinates $(\alpha_{t4}, \omega_{t4}, \theta, \phi)$ in radians and **must carry the $\texttt{-t4}$ suffix**. Never mix certificate scalars and $\mathbb{T}^4$ angles in the same formula.

In embodiments requiring non-convex admissible sets or periodic coordinate systems (variational quantum circuits, orbital mechanics), stability is analyzed in $\mathbb{T}^4$ coordinates. A projection Γ assigns $x_{t4}(E) \in \mathbb{T}^4$. Admissibility becomes $V_{t4}(E) \leq V_{\text{max}}$ where $V_{t4}(E) := d_{t4}(x_{t4}(E), x_{t4}^*)^2$ is the Lyapunov drift energy. This is the native mathematical structure of quantum circuit optimization landscapes and spacecraft attitude dynamics.

3.5 Why Three Axes Are Necessary

A single composite quality score collapses four structurally distinct failure states into the same representation:

Table 1: Same composite score; four distinct RSCT diagnoses and remediation paths.

α	κ_{gate}	σ	Scenario	RSCT Mode	Response
0.85	0.30	0.20	Clean signal; wrong solver geometry	1.4 Format Mismatch	RE.ENCODE
0.30	0.85	0.20	Dirty signal; solver filters noise	3.1 Fluent Hallucination	REPAIR
0.85	0.85	0.72	Compatible but dynamically unstable	2.1 Trajectory Divergence	BLOCK
0.30	0.30	0.20	Both axes weak; stable stall	4.1 Weakest-Link Cascade	REJECT

All four systems have identical composite score $q = 0.60$ under any prior binary framework. Under RSCT they are four entirely different failure conditions requiring four entirely different responses. This is not a constructed pathology. Row 2 describes the characteristic failure of frontier LLMs: high compatibility, low purity, confident hallucination. Row 3 describes variational quantum algorithms near barren plateaus: compatible but dynamically stalled. A single quality score detects neither.

The three-axis architecture is not a design preference. It is the minimal structure required to distinguish Table 1’s four rows before solver execution.

4 The Four Core Theorems

Capacity sets the ceiling.
Competence determines where you actually stand beneath it.
And a system can be perfectly stable and completely wrong — these are different axes.

This section proves four results that together justify the entire RSCT architecture. Each answers a different skeptical challenge:

1. **Theorem 4.1 (Factorization).** What does “intelligence” mean precisely, and how is it related to solver capacity and encoding competence?
2. **Theorems 4.2–4.4 (Independence).** Are $(\alpha, \kappa_{\text{gate}}, \sigma)$ genuinely independent dimensions, or are they correlated by a hidden constraint? If correlated, a composite score would suffice.
3. **Theorem 4.5 (Weakest-Link).** Is min-aggregation for κ_{gate} the only adversarially safe aggregation rule, or just one reasonable choice among many?

4.1 Theorem 1: Competence IS Intelligence

Definition 4.1 (Capacity). *The capacity of solver S on problem distribution $\mathcal{P}$ is the reciprocal of expected optimal difficulty:*

$$\text{cap}(S, \mathcal{P}) := \frac{1}{\mathbb{E}_{P \sim \mathcal{P}}[D^*(P, S)]}. \quad (8)$$

Capacity is a property of the solver and problem distribution alone. The encoding policy $\mathcal{M}$ does not appear. It is the ceiling, set before any encoding choices are made.

Definition 4.2 (Encoding Competence). *The competence of encoding policy $\mathcal{M}$ for solver S on distribution $\mathcal{P}$ is the expected fraction of the solver’s ceiling achieved by the encoding:*

$$\text{comp}(\mathcal{M}, S, \mathcal{P}) := \mathbb{E}_{P \sim \mathcal{P}} \left[\frac{D^*(P, S)}{D(P, \mathcal{M}(P), S)} \right] = \mathbb{E}_{P \sim \mathcal{P}} [\kappa(\mathcal{M}(P), S)]. \quad (9)$$

Competence lies in $[0, 1]$. At $\text{comp} = 1$ the encoding policy always achieves the solver’s theoretical optimum. At $\text{comp} = 0$ the encoding policy makes every problem maximally difficult for the solver.

Theorem 4.1 (Capacity–Competence Factorization). *Define intelligence as the reciprocal of expected task difficulty under encoding policy $\mathcal{M}$:*

$$\text{intelligence}(S, \mathcal{M}, \mathcal{P}) := \frac{1}{\mathbb{E}_{P \sim \mathcal{P}}[D(P, \mathcal{M}(P), S)]}. \quad (10)$$

Then:

$$\boxed{\text{intelligence}(S, \mathcal{M}, \mathcal{P}) = \text{cap}(S, \mathcal{P}) \times \text{comp}(\mathcal{M}, S, \mathcal{P})}. \quad (11)$$

Proof. Let $c = \text{cap}(S, \mathcal{P}) = 1/\mathbb{E}_P[D^*(P, S)]$. Define the *difficulty ratio* $\rho(P) := D(P, \mathcal{M}(P), S)/D^*(P, S)$. Competence is $\text{comp} = \mathbb{E}_P[1/\rho(P)]$. The factorization requires $\mathbb{E}_P[D^* \cdot \rho] = \mathbb{E}_P[D^*] \cdot \mathbb{E}_P[\rho]$, i.e. that optimal difficulty and the difficulty ratio are uncorrelated across $\mathcal{P}$.

Regularity condition. We assume $\text{Cov}_P(D^*(P, S), \rho(P)) = 0$. This holds whenever the encoding policy $\mathcal{M}$ is *problem-agnostic*: it does not systematically allocate encoding effort in proportion to intrinsic problem difficulty. Fixed prompting strategies, static RAG pipelines, and pre-trained projection heads all satisfy this condition. Difficulty-adaptive curricula (Section 6) may violate it; in that setting the factorization becomes a first-order approximation whose residual is bounded by the covariance $|\text{Cov}(D^*, \rho)|/\mathbb{E}[D^*]$.

Under the regularity condition:

$$\mathbb{E}_P[D(P, \mathcal{M}(P), S)] = \mathbb{E}_P[D^*(P, S) \cdot \rho(P)] = \mathbb{E}_P[D^*(P, S)] \cdot \mathbb{E}_P[\rho(P)] = \frac{1}{c} \cdot \frac{1}{\text{comp}(\mathcal{M}, S, \mathcal{P})}. \quad (12)$$

Taking the reciprocal: $\text{intelligence} = c \times \text{comp}$. □ □

Remark 4.1 (Scope of the Regularity Condition). *The uncorrelatedness assumption is structural, not incidental. It distinguishes two design regimes. In the fixed-policy regime (a deployed system with a frozen encoding pipeline), the assumption holds by construction and the factorization is exact. In the adaptive-policy regime (curriculum learning, active encoding search), the encoding policy co-evolves with problem difficulty, inducing non-zero $\text{Cov}(D^*, \rho)$. In that regime, the factorization remains a useful decomposition with a quantified residual, but it is no longer an identity. The RSSC formulation of Section 6 operates precisely at this boundary.*

How to read this result.

The formula $\text{intelligence} = \text{cap} \times \text{comp}$ is an algebraic identity, not a claim that capacity and competence are symmetric contributors. The crucial structural fact is:

- **Capacity is fixed.** Once the solver is chosen (a specific LLM, a VQE circuit, a compliance agent), cap is determined. It is a constant of the deployment context.
- **Competence is the variable.** $\text{comp} \in [0, 1]$ is entirely determined by the encoding policy $\mathcal{M}$. It is the single lever available to the system designer.
- **Intelligence is competence, scaled.** The equation says intelligence equals competence, with capacity serving as the scale factor for the ceiling being approached. Improving the encoding policy by a factor of k raises intelligence by exactly k , regardless of capacity.

The phrase “intelligence equals capacity times competence” can mislead if read as “increase either factor to increase intelligence.” Only competence is under the designer’s control at runtime. Capacity is the wall you are climbing toward.

Corollary 4.1.1 (Scaling Law Bound). *For a fixed encoding policy $\mathcal{M}$ with competence $\text{comp}_0 < 1$, increasing solver capacity by factor $\beta > 1$ raises intelligence by at most $\beta \cdot \text{comp}_0$. Improving the encoding policy from comp_0 to $\text{comp}_1 > \text{comp}_0$ yields multiplicative gain $\text{comp}_1/\text{comp}_0$, independent of capacity.*

Concrete consequence. If $\text{comp}_0 = 0.30$ (encoding policy achieves roughly a third of the solver’s theoretical ceiling), then doubling the solver raises intelligence to at most $2 \times 0.30 = 0.60$ of the new ceiling — a bound that grows only because the ceiling grew, not because the encoding improved. Improving encoding competence from 0.30 to 0.90 yields a $3\times$ gain from encoding work alone, which compounds with any capacity deployed.

When comp_0 is low, representation work dominates capacity scaling. The prevailing scaling paradigm has no formalization of encoding competence, and therefore no way to compute or escape this bound.

4.2 Theorems 2–4: The Three Axes Are Structurally Independent

If α , κ_{gate} , and σ were correlated by a hidden constraint — if, say, high purity automatically implied high compatibility — then a composite score would suffice and the three-gate architecture would be wasteful. The following theorems close this question by construction: we exhibit systems where the axes vary freely.

Theorem 4.2 (α – κ_{gate} Structural Independence). *Purity α and execution compatibility κ_{gate} are struc-*

structurally independent: the same purity can produce arbitrarily different compatibility, and the same compatibility can arise from arbitrarily different purity.

Proof by construction. Same α , different κ_{gate} : Consider a graph connectivity problem P . Two encodings E_1 (adjacency list) and E_2 (prose description) carry identical information: $R = 0.9, N = 0.1$, so $\alpha_1 = \alpha_2 = 0.9$. For a graph-traversal solver S : $D(P, E_1, S) \approx D^*(P, S)$, giving $\kappa_{\text{gate}} \approx 1$; but $D(P, E_2, S) \gg D^*(P, S)$ since S must parse graph structure from prose, giving $\kappa_{\text{gate}} \ll 1$. Equal purity; arbitrarily different compatibility.

Same κ_{gate} , different α : Let P be a robust classification problem. E_1 : $R = 0.8, N = 0, \alpha_1 = 1.0$ (clean encoding). E_2 : $R = 0.4, N = 0.4, \alpha_2 = 0.5$ (same features, half replaced with adversarial noise). If S is a robust classifier designed to filter perturbations, then $D(P, E_1, S) \approx D(P, E_2, S) \approx D^*(P, S)$, so $\kappa_{\text{gate}} \approx 1$ for both. Dramatically different purity; identical compatibility. $\square$ $\square$

Remark 4.2 (The Library Analogy). *The graph example crystallizes a deep structural point. The issue is never purely whether the information is present but whether it is structured to be accessible to the solver. A library is useless to someone who cannot read the language the books are written in — not because the information is absent, but because the encoding is geometrically incompatible with the reader’s processing machinery.*

Theorem 4.3 ($\kappa_{\text{gate}}\text{--}\sigma$ Structural Independence). *Execution compatibility κ_{gate} and turbulence σ are structurally independent: high compatibility does not imply low turbulence, and low turbulence does not imply high compatibility.*

Proof by construction. High κ_{gate} , high σ : Consider a near-degenerate numerical optimization. The encoding is maximally compatible — it exposes problem structure perfectly, $\kappa_{\text{gate}} \approx 1$ — but places solver dynamics precisely at the stability boundary. Small perturbations cause oscillation rather than convergence. The encoding is excellent; the dynamics are unstable. $\sigma \approx \sigma_{\text{thr}}$.

Low κ_{gate} , low σ : Encode a sorting task as natural language for a general-purpose LLM. The solver generates stable, predictable, coherent output — $\sigma \approx 0$ by any measure. But sorting is architecturally incompatible with the LLM: $D(P, E, S) \gg D^*(P, S)$, so $\kappa_{\text{gate}} \ll 1$. The system is placid and wrong. $\square$ $\square$

The second example deserves emphasis. A system can be perfectly stable — smooth outputs, no oscillation, no divergence, no alarm signals from a monitoring system tracking only σ — while being deeply incompatible with the problem it is supposed to solve. This is why gates for κ_{gate} and σ are both necessary: stability monitoring alone is not sufficient.

Theorem 4.4 (Three-Way Structural Independence). *The certificate triple $(\alpha, \kappa_{\text{gate}}, \sigma)$ is three-way structurally independent. For any target values $(\alpha^*, \kappa^*, \sigma^*) \in [0, 1]^3$, there exists a constructible encoding-solver pair simultaneously achieving all three.*

Proof sketch. We construct three orthogonal degrees of freedom. Let $(D_\alpha, D_\kappa, D_\sigma)$ be three independent design parameters:

1. D_α (training-time pool partition): controls the $R : N$ ratio in the context, directly setting α . No change to solver architecture.
2. D_κ (solver hyperparameter k): controls the geometric alignment between the encoding’s feature space and the solver’s principal components. Directly sets κ_{gate} without changing context content.

3. D_σ (inference-time perturbation rate): controls the variance of κ_i scores across modalities. Directly sets σ without affecting content or alignment.

No step in adjusting D_j constrains any parameter used by $D_{j'}$ for $j \neq j'$. The achievable certificate space is the full unit cube $[0, 1]^3$. Full construction in Appendix A. □ □

Remark 4.3 (Architectural Consequence). *No composite score can substitute for three separate certificate checks. A weighted average of α , κ_{gate} , and $1 - \sigma$ will fail to detect any system that passes on two axes while failing catastrophically on a third. The independence theorems guarantee such systems exist for any weighting scheme. This rules out not just naive aggregation but any single-score framework as a substitute for the three-axis architecture.*

4.3 Theorem 5: Weakest-Link Gating Optimality

The gatekeeper computes $\kappa_{\text{gate}} = \min(\kappa_H, \kappa_L, \kappa_{\text{int}})$. The natural question: is the minimum excessively conservative? Could a weighted average or harmonic mean preserve safety while avoiding false rejections?

Theorem 4.5 (Weakest-Link Gating Optimality). *For a compositional encoding with modal health scores $\{\kappa_1, \dots, \kappa_M\}$, κ_{gate} provides optimal resistance to masking and cascade failure under zero-trust enforcement if and only if $\kappa_{\text{gate}} = \min_i(\kappa_i)$.*

Proof sketch. Sufficiency: If $\kappa_{\text{gate}} = \min_i \kappa_i \geq \kappa_{\text{req}}$, then every individual modality satisfies the admissibility threshold. No single-modality failure can undercut the required compatibility level; cascade failure across the pipeline is impossible.

Necessity (adversarial bypass): Suppose aggregation rule f satisfies $f(\kappa_1, \dots, \kappa_n) > \min_i \kappa_i = \kappa_{i^*}$ for some encoding. The gatekeeper passes the encoding. An adversary who knows f targets modality i^* : degrades it below κ_{req} while the remaining modalities compensate to keep f above κ_{req} . The pipeline executes on a compromised encoding and fails downstream. The gatekeeper was successfully bypassed. Therefore any rule $f > \min$ violates zero-trust enforcement. □ □

NEVER replace $\min()$ with $\text{mean}()$ for κ_{gate} . The necessity proof rules out not just careless aggregation but *any* aggregation rule more permissive than the minimum, including principled alternatives like the harmonic mean or Bayesian model averaging. The argument is not that these are bad engineering choices; it is that they *cannot* provide the zero-trust security guarantee. Any implementation that uses mean is not implementing RSCT.

Summary of the four theorems. The factorization tells you what you are optimizing (encoding competence, the single controllable lever). The independence results tell you that all three axes are required to do it. The weakest-link result tells you how to aggregate modal health scores in the only way that is adversarially safe. The gatekeeper of Section 5 is the necessary enforcement consequence of these four results.

5 The RSCT Gatekeeper: Enforcement Architecture

*The time to catch a bad encoding is before the solver has seen it.
Gate ordering is a correctness guarantee under adversarial conditions,
not an efficiency heuristic.*

5.1 Design Rationale

The gatekeeper translates the three certificate axes into four sequential enforcement decisions. Each gate catches a failure mode the others cannot see:

- **Gate 1** catches noise saturation. When $N \geq 0.5$, Fano’s inequality (Proposition 2.2) proves that no solver can recover correct inference. No downstream gate should execute on noise-saturated input.
- **Gate 2** catches consensus failure. When sub-model interpretations are in phasor conflict ($c < c_{\min}$), the system is structurally incoherent regardless of average quality.
- **Gate 3** catches geometric incompatibility and turbulence. The Oobleck principle provides a dynamic threshold that stiffens as turbulence rises, enforcing the admissibility boundary $\sigma \leq \sigma_{\text{thr}}$ under Lemma 3.1.
- **Gate 4** catches low-level grounding failure. A low- κ_L encoding can pass all prior gates while being physically ungrounded.

5.2 The Oobleck Principle (Gate 3 Threshold)

Shear-thickening fluids solidify under high-velocity impact. RSCT translates this into computational control:

$$\kappa_{\text{req}}(\sigma) = \kappa_{\text{base}} + \lambda \cdot \sigma, \quad \kappa_{\text{base}} = 0.5, \quad \lambda = 0.4. \quad (13)$$

At low turbulence ($\sigma \approx 0$), the threshold is permissive: $\kappa_{\text{req}} \approx 0.5$. As the encoding becomes turbulent, the threshold stiffens: at $\sigma = 0.5$, $\kappa_{\text{req}} = 0.7$. The encoding must be more compatible, not equally compatible, to be admitted under stress.

A **Landauer Tolerance** buffer $[\kappa_{\text{req}} - \varepsilon_L, \kappa_{\text{req}})$ with $\varepsilon_L = 0.05$ prevents false rejections from measurement noise. Turbulence enforcement is two-tiered: (i) a hard boundary $\sigma > \sigma_{\text{thr}}$ forces RE._ENCODE regardless of κ_{gate} (enforcing Definition 3.5); (ii) in the Landauer gray zone ($\kappa_{\text{req}} - \varepsilon_L \leq \kappa_{\text{gate}} < \kappa_{\text{req}}$), a lower threshold $\sigma > 0.4$ acts as a deterministic tie-breaker, providing early warning before the hard boundary is reached.

5.3 Formal Algorithm

Algorithm 1 RSCT Pre-Execution Gatekeeper

Require: Encoding E , solver S , problem P

Require: Modal health $\{\kappa_1, \dots, \kappa_M\}$; YRSN simplex (R, S_{sup}, N) ; coherence c ; turbulence σ

Require: Thresholds: $\kappa_{\text{base}} = 0.5$, $\lambda = 0.4$, $\varepsilon_L = 0.05$, $c_{\text{min}} = 0.4$, $\kappa_L^{\text{min}} = 0.3$, $\sigma_{\text{thr}} = 0.5$

Ensure: Decision $\in \{\text{EXECUTE}, \text{REJECT}, \text{BLOCK}, \text{RE_ENCODE}, \text{REPAIR}\}$

1: **Gate 1 — Integrity Guard**

2: **if** $N \geq 0.5$ **then**

3: **return** $\langle \text{REJECT}, \text{“Noise saturation: } N \text{”} \rangle$

▷ Fano bound: correction impossible

4: **end if**

5: **Gate 2 — Consensus Gate**

6: **if** $c < c_{\text{min}}$ **then**

7: **return** $\langle \text{BLOCK}, \text{“Phasor conflict: } c \text{”} \rangle$

▷ Sub-models structurally incoherent

8: **end if**

9: **Gate 3 — Admissibility Gate (Oobleck)**

10: $\kappa_{\text{gate}} \leftarrow \min(\kappa_H, \kappa_L, \kappa_{\text{int}})$

▷ Theorem 4.5: min only

11: $\kappa_{\text{req}} \leftarrow \kappa_{\text{base}} + \lambda\sigma$

12: **if** $\sigma > \sigma_{\text{thr}}$ **then**

13: **return** $\langle \text{RE_ENCODE}, \text{“Turbulence exceeds } \sigma_{\text{thr}} \text{”} \rangle$

▷ Definition 3.5: outside $\mathcal{E}_{\text{adm}}$

14: **else if** $\kappa_{\text{gate}} < \kappa_{\text{req}} - \varepsilon_L$ **then**

15: **return** $\langle \text{RE_ENCODE}, \kappa_{\text{req}} - \kappa_{\text{gate}} \rangle$

▷ Hard incompatibility deficit

16: **else if** $\kappa_{\text{gate}} < \kappa_{\text{req}}$ **then**

▷ Landauer gray zone

17: **if** $\sigma > 0.4$ **then**

18: **return** $\langle \text{RE_ENCODE}, \text{“Turbulent gray zone”} \rangle$

19: **end if**

20: **end if**

21: **Gate 4 — Grounding Repair**

22: **if** $\kappa_L < \kappa_L^{\text{min}}$ **then**

23: **return** $\langle \text{REPAIR}, \kappa_L \rangle$

▷ Low-level physical signal ungrounded

24: **end if**

25: **return** $\langle \text{EXECUTE}, \kappa_{\text{gate}}, \sigma \rangle$

Proposition 5.1 (Gatekeeper Complexity). *Algorithm 1 runs in $O(M)$ time where M is the number of modal sources. No gate requires interaction with S or P , making it substrate-independent.*

NEVER reorder the gates. Executing Gate 3 before Gate 1 allows adversarially crafted encodings to produce arbitrarily high κ_{gate} estimates on noise-saturated input. Executing Gate 2 before Gate 1 allows consensus-failure decisions on input that should have been rejected outright. Gate ordering is a correctness guarantee under adversarial conditions, not an efficiency preference. Treating any gate as optional or softening its threshold without domain-specific calibration evidence violates the

zero-trust enforcement guarantee of Theorem 4.5.

5.4 Decision Reference

Table 2: Gatekeeper decision space and corresponding remediation actions.

Decision	Gate	Certificate condition	Remediation
REJECT	Gate 1	$N \geq 0.5$	Prune noise from G_R ; do not invoke solver
BLOCK	Gate 2	$c < c_{\min}$	Insert verification cycle; resolve sub-model conflict
RE_ENCODE	Gate 3	$\kappa_{\text{gate}} < \kappa_{\text{req}}(\sigma)$	Ensemble morphing; normalize turbulent G_R nodes
REPAIR	Gate 4	$\kappa_L < 0.3$	Expand G_R with higher-quality modal sources
EXECUTE	Clear	All gates pass	Invoke solver via <code>impl</code>

5.5 Worked Example: Two Gatekeeper Traces

Trace 1: Hard REJECT at Gate 1 (AML, Noise Saturation)

An AML pipeline processes a corporate account’s transaction network. Entity resolution has partially failed; spurious cross-entity edges corrupt the graph.

Input certificates: $R = 0.30$, $S_{\text{sup}} = 0.15$, $N = 0.55 \Rightarrow \alpha = 0.353$; $\kappa_H = 0.78$, $\kappa_L = 0.65$, $\kappa_{\text{int}} = 0.71 \Rightarrow \kappa_{\text{gate}} = 0.65$; $\omega = 0.70$, $c = 0.82$, $\sigma = 0.38$.

Gate 1: $N = 0.55 \geq 0.50 \Rightarrow \text{REJECT}$.

Without Gate 1, this encoding reaches Gate 3 with $\kappa_{\text{req}}(0.38) = 0.652$. Since $\kappa_{\text{gate}} = 0.65$ falls in the Landauer buffer $[0.602, 0.652)$ and $\sigma = 0.38 < 0.40$, the gray-zone tie-breaker would *pass* the encoding. A GNN would execute on a fabricated graph topology. Gate 1 blocks this regardless of all downstream certificate values.

Trace 2: Gray-Zone Gate 3, Turbulence as Tie-Breaker (CECL)

A CECL reserve pipeline processes a commercial real-estate cohort during rapidly rising interest rates. The macro scenario is near the validated distribution boundary.

Input certificates: $R = 0.72$, $S_{\text{sup}} = 0.18$, $N = 0.10 \Rightarrow \alpha = 0.878$; $\omega = 0.55 \Rightarrow \alpha_{\omega} = 0.708$; $\kappa_H = 0.81$, $\kappa_L = 0.76$, $\kappa_{\text{int}} = 0.74 \Rightarrow \kappa_{\text{gate}} = 0.74$; $c = 0.87$, $\sigma = 0.54$.

Gates 1 and 2 pass. Gate 3: $\sigma = 0.54 > \sigma_{\text{thr}} = 0.50 \Rightarrow \text{RE_ENCODE}$ (turbulence exceeds admissibility boundary, Definition 3.5).

Counterfactual: if $\sigma = 0.44$, then the turbulence hard-check passes. $\kappa_{\text{req}}(0.44) = 0.676$ and $\kappa_{\text{gate}} = 0.74$ clears Gate 3 outright. The RE_ENCODE decision is sensitive to the combination of marginal compatibility and elevated turbulence — exactly the behavior Oobleck is designed to enforce.

6 Theoretical Connection: Learning as Representation Search

The agent is not searching for the right action.

It is searching for the right encoding — the representational state from which correct actions become reachable and stable.

6.1 RSSC: A Reframing of Learning Dynamics

Remark 6.1 (Theoretical Reframing, Not Implementation). *This section presents RSSC as a theoretical lens for understanding learning systems — including, but not limited to, reinforcement learning. The DGM operational instantiation (Appendix B) does **not** implement online learning of any kind. DGM uses:*

- Fixed embeddings computed once from pre-trained encoders
- Geometric compatibility via deterministic cosine similarity
- Structural graph morphing (not weight updates)
- Zero gradient descent at runtime

Learning in YRSN occurs offline during R/S/N projection head training (Section 2), not during multi-agent execution.

Under RSCT, the environment state is not the world — it is the encoding E_t . What the agent controls is not which action to take but how the problem is represented to the solver. Actions are representational transformations $A : E_t \rightarrow E_{t+1}$. The optimization is:

$$\max_{\pi} \mathbb{E}_{\pi}[\kappa_{\text{gate}}(E_T, S)] \quad \text{subject to} \quad \sigma(E_t, S) \leq \sigma_{\text{thr}} \quad \forall t. \quad (14)$$

The stability constraint is a safety condition, not merely an efficiency preference. A trajectory that violates $\sigma \leq \sigma_{\text{thr}}$ at intermediate steps is dynamically unsafe: it risks the collapse modes of type 2.x (trajectory divergence, gradient starvation). Correct actions only become reachable and stable from encodings that lie within $\mathcal{E}_{\text{adm}}$. An agent that arrives at a compatible encoding via an unstable path cannot be trusted to stay there.

This formulation has an equivalent Lagrangian relaxation for policy gradient methods:

$$\max_{\pi} \mathbb{E}_{\pi}[\kappa_{\text{gate}}(E_T, S)] - \mu \sum_{t=0}^T \max(0, \sigma(E_t, S) - \sigma_{\text{thr}}). \quad (15)$$

The penalty form is more tractable for training; the constraint form is stronger for safety-critical deployment.

6.2 Curriculum Learning as Stable Region Expansion

Under RSSC, a curriculum is the progressive expansion of the stable representational region $\mathcal{E}_{\text{adm}}(S_t, \mathcal{P}_t)$ as solver capability develops. Optimal training feeds solvers encodings near $\sigma \approx \sigma_{\text{thr}}$, forcing acquisition of representational competence to push encodings back into the stable regime.

Phase boundaries are not universal constants. They are distribution-dependent parameters that must be empirically calibrated per encoder and domain. The Curriculum Phase Configuration is:

Definition 6.1 (Curriculum Phase Configuration). *Phase $\Phi \in \{PHASE_1, PHASE_2, PHASE_3\}$ is characterized by a 4-tuple $\text{Config}(\Phi) = (\kappa_{\text{floor}}, \sigma_{\text{ceil}}, \kappa_{\text{base}}, \sigma_{\text{thr}})$ where κ_{floor} is the minimum κ_{gate} required for Φ , σ_{ceil} is the maximum σ tolerated, and the remaining two govern Gate 3’s Oobleck threshold.*

Remark 6.2 (Progressive Constraint Relaxation). *Phases relax constraints monotonically:*

$$\kappa_{\text{floor}}^{(1)} > \kappa_{\text{floor}}^{(2)} > \kappa_{\text{floor}}^{(3)}, \quad \sigma_{\text{ceil}}^{(1)} < \sigma_{\text{ceil}}^{(2)} < \sigma_{\text{ceil}}^{(3)}. \quad (16)$$

Solvers master simple, stable encodings before advancing to complex, turbulent representations. Phase 3 tolerates higher turbulence because synthesis tasks require integrating divergent evidence sources; Phase 1 demands lower turbulence because foundational encoding must be clean for competence to develop stably.

Clarification: *This mastery occurs during offline training of projection heads and embedding models, not during DGM runtime execution. At runtime, DGM validates that graphs match their assigned curriculum phase via fixed threshold checks ($\sigma \leq \sigma_{\text{ceil}}$, $\kappa_{\text{gate}} \geq \kappa_{\text{floor}}$), with no weight updates.*

Remark 6.3 (Calibration Requirement). *Phase boundaries must be recalibrated whenever the embedding model changes, embedding dimensionality changes, or the domain shifts. The recommended 2-sigma calibration rule: given $n \geq 40$ observations $\{\kappa_{\text{gate}}^{(i)}\}$ from phase Φ , set $\kappa_{\text{floor}}^* = \max\{0.3, \mu_{\kappa} - 2\sigma_{\kappa}\}$, which captures $\approx 95\%$ of the empirical distribution while maintaining a safety margin. Empirical validation (S7.003) confirmed: SentenceTransformer embeddings exhibit $\approx 3\times$ higher κ_{gate} variance than hash encodings, requiring threshold reduction from 0.48 to 0.46 to prevent spurious curriculum rejections.*

7 Substrate-Independence: RSCT Across Deployment Domains

The physics changes. The certificate structure does not.

Whether the solver is a transformer, a quantum circuit, or a memristor array, the encoding either lies within the admissible region or it does not.

7.1 The Substrate-Independence Claim

The independence theorems of Section 4 make a structural claim: the certificate triple $(\alpha, \kappa_{\text{gate}}, \sigma)$ spans a genuinely three-dimensional risk space that is solver-agnostic. A failure mode characterized as Mode 3.1 (Fluent Hallucination) has identical certificate structure — high κ_{gate} , low α — whether the solver is a transformer generating a regulatory filing or a NeRF renderer generating a 3D scene. The phenomenology differs; the encoding failure does not.

Table 3: RSCT triad and dominant collapse mode across deployment domains. In every case the binding failure is encoding-induced, not solver-capacity-limited.

Domain	Problem (P)	Encoding (E)	κ Proxy	Collapse Mode
Language (LLM)	User query	Prompt + RAG context	Answer quality	3.1 Fluent Hall.
Quantum VQA	Hamiltonian	Ansatz circuit	Gradient variance	2.2 Grad. Starv.
Neuromorphic	Pattern matching	Analog voltage spikes	Filament integrity	2.4 Substrate Deg.
Spatial AI (NeRF)	3D reconstruction	Radiance + photomets.	View synth. PSNR	3.1 Fluent Hall.
Orbital Mfg.	Microgravity fab.	Telemetry schema	Wafer yield	4.1 Weakest-Link
Financial Compl.	SR 11-7 audit	Retrieved guidance	SR 11-7 pass rate	3.1, 4.4

The appearance of Mode 3.1 in both spatial AI and language models is not coincidence. Any system that generates outputs without a grounded cross-modal integrity check between its encoding and a structured reference source will produce confident, fluent, wrong outputs when the encoding is noise-contaminated. The certificate failure — high κ_{gate} with low α — is identical. A unified theory of representation-solver compatibility predicts exactly this.

7.2 The Tribes of Machine Learning, Recontextualized

Table 4: Each ML paradigm is a local inhabitant of a specific compatibility regime. Paradigm failures are exits from $\mathcal{E}_{\text{adm}}$, not solver deficiencies.

Tribe	High- κ regime	Characteristic collapse	RSCT exit condition
Symbolists	Combinatorial constraints	State-space explosion	$\sigma \gg \sigma_{\text{thr}}$ (Mode 2.1)
Connectionists	High-dim. perception	OOD hallucination	$\alpha \rightarrow 0$ outside train dist. (Mode 3.4)
Bayesians	Uncertainty quantif.	Intractable margin.	Prior misspec. (Mode 3.3)
Evolutionaries	Non-differentiable land.	Deceptive fitness	$\sigma \gg \sigma_{\text{thr}}$, barren plateaus
Analogizers	Margin maximization	Dimensionality curse	$\kappa_{\text{gate}} \rightarrow 0$ in high- d
Navigators	Heterogeneous pipelines	Gatekeeper miscalibration	Proxy-estimation failure

Each paradigm found a real effect and mistook it for a universal theory of intelligence. The boundary between paradigms is not a boundary between competing theories — it is a boundary between compatibility regimes. RSCT provides the unified coordinate system in which all five tribes are local specialists.

The five-paradigm taxonomy originates with Domingos, who has recently proposed *tensor logic* [12] as a unifying *language* for the tribes — reducing neural, symbolic, and statistical computation to a single tensor equation construct. RSCT offers a complementary unification: not a common language but a common *certification geometry* that diagnoses why each tribe fails outside its native compatibility regime and what gatekeeper enforcement would prevent the failure.

The Navigator class represents the logical endpoint of this reorientation: a system whose primary function is not to solve problems directly but to certify and maintain the representational conditions under which any solver can operate reliably.

8 Discussion

8.1 Why Scaling Alone Is Bounded

The prevailing AI development paradigm is internally consistent under the intrinsic difficulty assumption: if difficulty is a property of P alone, more solver capacity is always a valid response to underperformance. Corollary 4.1.1 is incompatible with this framing.

Under RSCT, every performance gain from capacity scaling is bounded above by $\beta \cdot \text{comp}_0$. That bound can be made arbitrarily small by holding comp_0 fixed, and the bound is available to be computed because RSCT provides a formalization of encoding competence. The scaling paradigm does not have this formalization, and therefore cannot compute or escape the bound.

Scale without certification is just expensive noise.

8.2 Historical Context: Five Convergent Traditions

RSCT did not arise in a vacuum. Its three certification axes — signal purity α , geometric compatibility κ_{gate} , and representational turbulence σ — each descend from a mature intellectual tradition that developed independently over decades. Two further traditions — representation learning and neuromorphic computing — supplied the central thesis (encoding is the binding variable) and the physical evidence (intelligence is interface-governed, not capacity-governed) respectively. What is new is not any individual ingredient but their formal combination into a pre-execution certification architecture. Tracing these five lineages clarifies both why the framework takes its present form and why this combination did not emerge earlier.

The statistical tradition (α axis). The distinction between signal and noise is the founding act of mathematical statistics. Fisher’s sufficiency (1922), Neyman–Pearson hypothesis testing (1933), and Tukey’s exploratory data analysis (1977) each formalized different aspects of the same question: *what fraction of observed variation carries information about the quantity of interest?* The YRSN decomposition (Section 2) inherits this question directly, and the α purity measure is grounded in Fano’s inequality (1961) — a channel-coding result that bounds recoverability of a transmitted signal given observed noise. The statistical tradition supplied the signal-purity axis but never asked whether the purified signal was *geometrically compatible* with any particular solver.

The computational tradition (κ axis). Computational learning theory — Valiant’s PAC framework (1984), Vapnik’s VC dimension and structural risk minimization (1995), and the subsequent neural scaling laws of Kaplan et al. (2020) and Hoffmann et al. (2022) — formalized the relationship between solver capacity and generalization. These results treat the representation as a fixed input and ask how much capacity the solver needs. RSCT inverts this question: given a fixed solver, what fraction of its capacity does the encoding actually unlock? The compatibility coefficient κ_{gate} is not a capacity bound; it is a *utilization* measure. The computational tradition supplied the capacity axis but treated encoding quality as exogenous.

The dynamical-systems tradition (σ axis). Lyapunov stability theory (1892), Khalil’s nonlinear-systems framework [3], and the control-theoretic study of robustness margins formalized when a dynamical system remains in a safe operating region under perturbation. The representational turbulence σ and its $\mathbb{T}^4$ stability representation (Section 3) import this machinery into the encoding–solver interface: turbulence measures how rapidly the compatibility landscape shifts under small perturbations of the encoding, and the gatekeeper’s $\sigma \leq \sigma_{\text{thr}}$ constraint is a stability margin. The dynamical-systems tradition supplied the stability axis but had no notion of representational purity or solver compatibility to stabilize.

The representation-learning tradition (the encoding question). The explicit claim that *representation quality determines learning success* crystallized with Bengio, Courville, and Vincent’s [5] influential review, which catalogued desiderata for good representations — smoothness, sparsity, temporal coherence — without providing any formal certification mechanism. The subsequent deep-learning revolution validated this intuition at industrial scale: every architectural advance from ResNets to transformers can be read as an implicit bet on a particular encoding geometry. Yet the field’s response to encoding failure (hallucination, distributional shift, reward hacking) remained post-hoc: detect the symptom, patch the model, retrain. The representation-learning tradition supplied the central *thesis* — encoding is the binding variable — but never supplied the *certification architecture* that could enforce encoding quality before execution.

The neuromorphic tradition (intelligence at the substrate). Chua’s postulation of the memristor [16] (1971) as the fourth fundamental circuit element, and its experimental realization by Strukov et al. [17] at HP Labs (2008), opened an alternative path to intelligent computation: one in which the encoding *is* the physical substrate. In a memristor crossbar array, resistance states are the encoding, the crossbar topology is the solver, and filament degradation is substrate-level turbulence (Mode 2.4 in Section 7). Intelligence emerges not from scaling transistor counts but from the compatibility between analog state configurations and the computational geometry they must support. The neuromorphic program — from Chua’s edge-of-chaos dynamics through Intel’s Loihi and IBM’s TrueNorth to the current wave of in-memory computing architectures — demonstrates that milliwatt-scale systems can match or exceed the per-task performance of datacenter-scale deployments when the encoding–solver interface is well-matched. Even the conventional GPU ecosystem has begun to internalize this lesson: NVIDIA Research itself now argues that small language models, not large ones, are the future of agentic AI — contending that “capability — not the parameter count — is the binding constraint” [18]. Their proposed LLM-to-SLM conversion algorithm relies on ad hoc task clustering and fine-tuning but lacks any formal certification of when a smaller model’s encoding–solver interface is admissible. The industry’s pivot from training-scale to inference-efficiency, and from datacenter to edge, reflects a growing recognition that raw capacity alone cannot compensate for encoding incompatibility when the compute budget is physically constrained. The neuromorphic tradition supplied the *physical proof* that intelligence is interface-governed, not capacity-governed — but had no formal language for certifying when that interface is admissible.

Why convergence required a new architecture. Each tradition solved its own problem completely yet left the others unformalized. Statisticians could certify signal purity without knowing whether the solver could exploit it. Learning theorists could bound capacity without measuring whether the encoding utilized it. Control theorists could enforce stability without specifying what representational quantity was being stabilized. Representation-learning researchers could articulate the thesis that encoding matters without any mechanism to enforce it. And neuromorphic engineers could build systems where encoding *is* the substrate without any formal guarantee that the substrate state was admissible. The structural independence theorems (Theorems 4.2–4.4) formalize this historical observation: *no tradition’s axis constrains either of the others*, which is precisely why decades of progress within each stream did not produce a unified certification framework. RSCT’s contribution is the architecture that combines all three into a single enforceable pre-execution certificate — an architecture whose necessity the independence theorems guarantee.

8.3 Limitations and Open Problems

Proxy heuristics. Neither $\kappa(E, S)$ nor $\sigma(E, S)$ is directly observable at runtime without solving the problem. The gatekeeper operates on proxy estimators $\hat{\kappa}$ and $\hat{\sigma}$. The gap between proxy and true value is an open research problem.

Calibration of λ . The default $\lambda = 0.4$ for the Oobleck coefficient requires domain-specific calibration. Cross-domain benchmarking of false-rejection rate vs. cascade-failure rate as a function of λ is a priority.

Compositionality of competence in multi-agent chains. Theorem 4.1 treats competence as an expectation over a single problem class. Formalizing the multiplicative propagation of gatekeeper errors across an n -stage agentic pipeline is an open problem with direct implications for long-horizon AI safety. If gatekeeper accuracy per stage is p , a naive chain of n agents has pipeline reliability $\leq p^n$, motivating both improved proxy accuracy and redundant certification stages.

Adversarial robustness of proxies. If proxy estimators $\hat{\kappa}$ and $\hat{\sigma}$ can be adversarially manipulated independently of the encoding they are estimating, the gatekeeper’s security guarantees break. Formal adversarial robustness analysis of the proxy-estimation layer is a critical open problem, particularly for financial compliance and physical hardware deployment.

9 Conclusion and Future Research

*Solvability is not a property of problems.
It is engineered into the encoding —
and the encoding is the variable the field forgot to measure.*

The central claim of RSCT is that solvability is representation-governed. Difficulty is not an intrinsic property of a problem instance P alone, but a triadic functional $D(P, E, S)$ that depends on the representation (or encoding) E and the solver S . Stable execution is therefore governed not by raw capacity in isolation, but by whether the encoding lies within an admissible region $E_{\text{adm}}(S, P)$ determined by compatibility and turbulence constraints.

This shift provides a single structural lens for failures that are often treated as unrelated: LLM hallucination, quantum barren-plateau collapse, analog substrate drift, and high-stakes decision errors in applied domains. Under RSCT, these phenomena can be understood as instances of trajectories exiting admissibility:

the operational breakdown is expressed as a loss of representation–solver compatibility or a rise in representational turbulence, even when downstream symptoms differ.

RSCT operationalizes this view through five contributions. First, we formalize signal purity via the normalized decomposition (R, S_{sup}, N) and derive a reliability-weighted quality α , a reliability coefficient ω , and a temperature-like control parameter τ (Section 2). Second, we introduce geometric compatibility and turbulence as independent axes that jointly determine execution admissibility, including a tripartite compatibility architecture and a stability representation (Section 3). Third, we prove that these axes do not collapse into a single scalar risk proxy and therefore cannot be safely replaced by confidence alone (Section 4). Fourth, we derive the RSCT gatekeeper as the enforcement consequence of these structural facts, yielding a pre-execution control mechanism that implements typed decisions rather than binary accept/reject (Section 5). Fifth, we connect static certification to dynamic learning by reframing learning dynamics as representation search under stability constraints (RSSC), clarifying where learning occurs and why execution-time governance remains auditable (Section 6).

The resulting architecture is substrate-independent by construction: the certificate space $\langle \alpha, \kappa_{\text{gate}}, \sigma \rangle$ expresses the same structural risk dimensions whether the solver is a transformer, a graph neural network, a variational quantum circuit, or a control system operating near physical limits. In this sense, RSCT is not merely a catalog of failure cases; it specifies a compatibility state space and an enforcement interface that can be evaluated and acted upon prior to execution.

The limitations and open problems discussed in Section 8.3 clarify the boundary of the present contribution and motivate several promising research directions.

Future research. We highlight four directions that follow naturally from the framework. (1) *Cross-domain calibration*: empirically calibrate the dynamic turbulence threshold σ_{thr} across on the order of 10^3 domains to study transferable phase boundaries and to support oracle-free diagnostic procedures. (2) *Autonomous coordination*: extend RSCT governance to multi-agent topologies, including deterministic, closed-loop certificate exchange and morphism-driven routing, and evaluate when such mechanisms improve stability and performance relative to open-ended action search under explicit safety constraints. (3) *Hardware and hybrid substrates*: develop hardware-accelerated implementations for analog memristor arrays and hybrid quantum environments, where RSCT-style gates provide operational checks for turbulence-induced degradation and formal conditions relevant to drift, instability, and barren-plateau-related failure modes. (4) *Topological verification*: explore topological extensions based on homological zero-knowledge proofs to enable privacy-preserving structural validation of certified artifacts without revealing sensitive internal structure.

Taken together, these directions address a core challenge in modern scaling regimes: increased capacity yields diminishing returns when representation quality is the limiting factor. RSCT supplies both a diagnostic state $\langle \alpha, \kappa_{\text{gate}}, \sigma \rangle$ and a deterministic enforcement interface that makes scaling more predictable, controllable, and safe.

Code Availability

We release a research prototype (work in progress) implementing the RSCT gatekeeper and related components, including a prototype DGM operational instantiation, YRSN projection heads, and RSSC curriculum templates: github.com/NextShiftConsulting/swarm-it-adk. The codebase is evolving and intended primarily for transparency and replication of core mechanisms rather than as a production library;

interfaces, file layout, and default parameters may change. Where applicable, we provide versioned tags (or commit hashes), pinned dependency files, and example run scripts to reproduce the gatekeeper traces and DGM operational examples described in this paper.

Notation Summary

Symbol	Definition
P, E, S	Problem, Encoding, Solver
$D(P, E, S)$	Relational difficulty; $D^*(P, S) := \inf_E D(P, E, S)$
R, S_{sup}, N	Relevant signal, Superfluous content, Noise; $R + S_{\text{sup}} + N = 1$. S_{sup} never abbreviated to S .
α	Signal purity: $R/(R + N) \in [0, 1]$
ω	Distributional reliability (OOD confidence): $\omega \in [0, 1]$
α_ω	Reliability-weighted quality: $\omega\alpha + (1 - \omega)\alpha_{\text{base}}$
τ	Temperature: $1/\alpha_\omega$. Scheduling instrument only; not in gates or theorems.
κ	Atomic compatibility: $D^*(P, S)/D(P, E, S) \in [0, 1]$
κ_{gate}	Execution score: $\min(\kappa_H, \kappa_L, \kappa_{\text{int}})$
$\kappa_H, \kappa_L, \kappa_{\text{int}}$	High-level health, low-level health, interface compatibility
$\kappa_A, \kappa_E, \kappa_T$	Diagnostic simplex: Algorithmic, Encoding, Tractability. Post-hoc only.
$\kappa_{\text{req}}(\sigma)$	Dynamic threshold: $\kappa_{\text{base}} + \lambda\sigma$; defaults $\kappa_{\text{base}} = 0.5, \lambda = 0.4$
σ	Turbulence: $\text{std}(\{\kappa_i\}_{i=1}^M) \in [0, 1]$
σ_{thr}	Turbulence threshold; default 0.5
c	Consensus coherence; $c_{\text{min}} = 0.4$
$\mathcal{E}_{\text{adm}}(S, P)$	Admissible encoding region
λ	Oobleck stiffening coefficient; default 0.4
$\text{cap}(S, \mathcal{P})$	Solver capacity: $1/\mathbb{E}_P[D^*(P, S)]$. Encoding policy does not appear.
$\text{comp}(\mathcal{M}, S, \mathcal{P})$	Encoding competence: $\mathbb{E}_P[\kappa(\mathcal{M}(P), S)] \in [0, 1]$
$\text{intelligence}(S, \mathcal{M}, \mathcal{P})$	Intelligence: $\text{cap} \times \text{comp} = \text{comp}$ scaled to capacity ceiling
$\mathbb{T}^4$	4-torus for stability coordinates. $(\alpha_{t4}, \omega_{t4}, \theta, \phi)$ must carry $\text{t}4$ suffix. Never mix with certificate scalars.

References

- [1] McGuire, G., Tugemann, B., & Civario, G. (2014). There is no 16-clue Sudoku. *Experimental Mathematics*, 23(2), 190–217.
- [2] McClean, J. R., et al. (2018). Barren plateaus in quantum neural network training landscapes. *Nature Communications*, 9, 4812.
- [3] Khalil, H. K. (2002). *Nonlinear Systems* (3rd ed.). Prentice Hall.
- [4] Lee, Y. S., Wetzels, E. D., & Wagner, N. J. (2003). The ballistic impact characteristics of Kevlar woven fabrics impregnated with a colloidal shear thickening fluid. *Journal of Materials Science*, 38(13), 2825–2833.

- [5] Bengio, Y., Courville, A., & Vincent, P. (2013). Representation learning: A review and new perspectives. *IEEE TPAMI*, 35(8), 1798–1828.
- [6] DiCarlo, J. J., & Cox, D. D. (2007). Untangling invariant object recognition. *Trends in Cognitive Sciences*, 11(8), 333–341.
- [7] Zou, A., et al. (2023). Representation engineering: A top-down approach to AI transparency. *arXiv:2310.01405*.
- [8] Tishby, N., Pereira, F. C., & Bialek, W. (2000). The information bottleneck method. *arXiv:physics/0004057*.
- [9] Wei, J., et al. (2022). Chain-of-thought prompting elicits reasoning in large language models. *NeurIPS*, 35, 24824–24837.
- [10] Trinh, T. H., et al. (2024). Solving olympiad geometry without human demonstrations. *Nature*, 625(7995), 476–482.
- [11] Lu, Y., et al. (2026). DyTopo: Dynamic topology routing for multi-agent reasoning via semantic matching. *arXiv:2602.06039*.
- [12] Domingos, P. (2025). Tensor logic: The language of AI. *arXiv:2510.12269*.
- [13] Arora, S., & Barak, B. (2009). *Computational Complexity: A Modern Approach*. Cambridge University Press.
- [14] Zhu, R.-J., et al. (2025). Scaling latent reasoning via looped language models. *arXiv:2510.25741*.
- [15] Martin, R. A. (2026). Technical Specification: Methods and Systems for Structured Compatibility Certification and Execution Control. *Unpublished Manuscript*.
- [16] Chua, L. O. (1971). Memristor—the missing circuit element. *IEEE Trans. Circuit Theory*, 18(5), 507–519.
- [17] Strukov, D. B., Snider, G. S., Stewart, D. R., & Williams, R. S. (2008). The missing memristor found. *Nature*, 453(7191), 80–83.
- [18] Belcak, P., Heinrich, G., Diao, S., Fu, Y., Dong, X., Muralidharan, S., Lin, Y. C., & Molchanov, P. (2025). Small language models are the future of agentic AI. *arXiv:2506.02153*.

A Independence Construction (Theorem 4.4)

We provide the full constructive proof of three-way structural independence (Theorem 4.4).

Setup. Let $D_\alpha \in [0, 1]$ be a pool partition parameter, $D_\kappa \in [0, 1]$ a rotation angle parameter, and $D_\sigma \in [0, 1]$ an inference-time perturbation rate. We construct a family of (E, S) pairs indexed by $(D_\alpha, D_\kappa, D_\sigma)$ such that:

$$\alpha(E) = D_\alpha, \quad \kappa(E, S) = f(D_\kappa), \quad \sigma(E, S) = D_\sigma, \quad (17)$$

where f is a monotone map with $f(D_\kappa) \in [0, 1]$.

Construction.

1. α via D_α : The context pool is partitioned into fraction D_α relevant tokens and $1 - D_\alpha$ adversarial noise tokens. This directly sets $\alpha = D_\alpha$. No solver parameter is changed.
2. κ via D_κ : The solver operates on a fixed feature space $\mathcal{X} \subset \mathbb{R}^d$. The encoding applies a rotation $R_\theta \in \text{SO}(d)$ with angle $\theta = D_\kappa \cdot \pi/2$ to the feature coordinates before presenting them to the solver. At $D_\kappa = 0$ the features are perfectly aligned with the solver's decision boundary ($\kappa \approx 1$); at $D_\kappa = 1$ the features are orthogonally rotated ($\kappa \approx 0$). The map $f(D_\kappa) = \cos(D_\kappa \cdot \pi/2)$ is monotonically decreasing. Crucially, rotation changes the geometric relationship between encoding and solver without altering context content or affecting the R/N ratio.
3. σ via D_σ : A fraction D_σ of the M modal sources is replaced with adversarial perturbations at inference time. By Lemma 3.1, this monotonically controls σ . No change to content or solver geometry.

Independence. The three parameters $(D_\alpha, D_\kappa, D_\sigma)$ are fully decoupled: adjusting any one does not constrain the others. A target point $(\alpha^*, \kappa^*, \sigma^*)$ is achieved by setting $D_\alpha = \alpha^*$, $D_\kappa = (2/\pi) \arccos(\kappa^*)$, $D_\sigma = \sigma^*$. The achievable set is the full unit cube $[0, 1]^3$. $\square$

B Dual Graph Morphing: Operational Instantiation of RSCT

B.1 The Relationship Between RSCT and DGM

The Dual Graph Morphing (DGM) system instantiates RSCT through:

- **Fixed embeddings:** Computed once via OmniEncoder, never updated
- **Geometric φ -morphism:** Deterministic cosine similarity, no trainable parameters
- **Derived certificates:** Statistical aggregation from φ -morphism pair scores
- **Deterministic gates:** Threshold comparisons (Algorithm 1)
- **Structural adaptation:** Graph topology changes via morph operators

DGM has **zero runtime learning**. No gradient descent, no policy networks, no reward signals, no exploration/exploitation. The "learning" discussed in Section 6 refers to the theoretical RSSC formulation and offline training of R/S/N projection heads, not to DGM's operational mechanism.

Multi-agent coordination in DGM is *compatibility-governed*, not policy-learned.

RSCT is a mathematical theory. The gatekeeper (Algorithm 1) is its enforcement specification. Dual Graph Morphing (DGM) is the operational system that *instantiates* RSCT at runtime: it constructs the two graphs over which the certificate triple $(\alpha, \kappa_{\text{gate}}, \sigma)$ is computed, generates the φ -morphism that produces κ_{int} , and executes the four morph operators that implement the gatekeeper's four non-EXECUTE decisions.

The key relationship is:

RSCT concept		DGM instantiation
Encoding E (representation space)	→	G_R : the Representation Graph
Solver S (solver space)	→	G_S : the Solver Graph
$\kappa(E, S) = D^*/D$	→	φ -morphism pair score
$\kappa_{\text{int}} = \min_i \kappa_i$	→	min over φ -map pairs
$\sigma = \text{std}(\{\kappa_i\})$	→	std over φ -map pair scores
Gate 1 REJECT → prune	→	<code>pruning_morph()</code>
Gate 2 BLOCK → verify	→	<code>verification_morph()</code>
Gate 3 RE_ENCODE → expand	→	<code>ensemble_morph()</code>
Gate 4 REPAIR → ground	→	<code>event_expansion_morph()</code>
Gate clear EXECUTE	→	<code>Fire AgentNode.impl()</code>

B.2 The Dual Graph Structure

Definition B.1 (Representation Graph G_R). $G_R(t) = (V_R, E_R)$ is a directed acyclic graph whose nodes are EventNodes, each carrying:

- A content embedding vector $\mathbf{e}_R \in \mathbb{R}^d$ (produced by the encoder from the node’s payload);
- A local YRSN certificate (R_i, S^i, N_i) satisfying the simplex constraint;
- A local turbulence estimate $\sigma_{\text{local},i}$;
- A modality tag $m_i \in \{\text{TEXT}, \text{IMAGE}, \text{AUDIO}, \text{VIDEO}, \text{STRUCT}, \text{EVIDENCE}\}$.

$G_R(t)$ is the operational instantiation of the encoding E in Definition 2.1. Its global certificate is derived by aggregating local node certificates.

Definition B.2 (Solver Graph G_S). $G_S(t) = (V_S, E_S)$ is a directed graph whose nodes are AgentNodes, each carrying:

- A capability centroid vector $\mathbf{e}_S \in \mathbb{R}^d$ (produced by the encoder from the agent’s role, description, and system prompt);
- A SolverContract specifying $(\kappa_{\text{base}}, \sigma_{\text{thr}}, c_{\text{min}})$ for this agent;
- An `impl` function that fires the underlying framework (CrewAI agent, A2A endpoint, Swarm-It worker) when the gatekeeper returns EXECUTE.

$G_S(t)$ is the operational instantiation of the solver S in Definition 3.1.

Definition B.3 (Compatibility Morphism φ). The compatibility morphism $\varphi : G_R \rightarrow G_S$ maps each EventNode $v_R \in V_R$ to the AgentNode $v_S \in V_S$ with maximum geometric compatibility:

$$\varphi(v_R) = \arg \max_{v_S \in V_S} \kappa(\mathbf{e}_R^{(v_R)}, \mathbf{e}_S^{(v_S)}), \quad (18)$$

where $\kappa(\mathbf{e}_R, \mathbf{e}_S)$ is the normalized cosine similarity between the two embedding vectors. The pair score $\kappa_i = \kappa(\mathbf{e}_R^{(v_i)}, \mathbf{e}_S^{(\varphi(v_i))})$ for each event-agent pair is the atomic compatibility of Definition 3.2, operationalized.

The interface compatibility κ_{int} then follows directly from Theorem 4.5:

$$\kappa_{\text{int}} = \min_{v_R \in V_R} \kappa(\mathbf{e}_R^{(v_R)}, \mathbf{e}_S^{(\varphi(v_R))}). \quad (19)$$

And the turbulence σ is the standard deviation of those same pair scores:

$$\sigma = \text{std}\left(\left\{\kappa(\mathbf{e}_R^{(v_R)}, \mathbf{e}_S^{(\varphi(v_R))})\right\}_{v_R \in V_R}\right). \quad (20)$$

This is the formal grounding of the turbulence definition in Definition 3.5: σ is the scatter of φ -morphism pair scores, not an independent measurement.

B.3 Global Certificate Derivation

Given G_R and G_S , the global RSCT certificate is computed in three stages:

1. **YRSN aggregation.** The global (R, S_{sup}, N) simplex is computed as a weighted mean over active EventNode local certificates, with weights proportional to node confidence:

$$N_{\text{global}} = \frac{\sum_{v \in V_R} w_v \cdot N_v}{\sum_v w_v}, \quad (21)$$

and similarly for $R_{\text{global}}, S_{\text{sup, global}}$. The global purity $\alpha = R_{\text{global}} / (R_{\text{global}} + N_{\text{global}})$.

2. **Modal health.** High-level health κ_H is the mean φ -pair score over TEXT and STRUCT nodes; low-level health κ_L is the *minimum* φ -pair score over IMAGE, AUDIO, VIDEO, and EVIDENCE nodes. Both follow from Definition 3.3.
3. **Execution score.** $\kappa_{\text{gate}} = \min(\kappa_H, \kappa_L, \kappa_{\text{int}})$, as required by Theorem 4.5. Coherence c is computed as the fraction of φ -pairs with $\kappa_i > 0.5$.

This triple $(\alpha, \kappa_{\text{gate}}, \sigma)$ with coherence c is the input to Algorithm 1.

B.4 The Four Morph Operators

Each gatekeeper non-EXECUTE decision maps to a morph operator that modifies (G_R, G_S) to repair the certificate deficiency. The operators are designed so that each addresses *only* the failure mode that triggered it, leaving the other graph unchanged where possible.

Table 5: Morph operator correspondence: gatekeeper decision $\rightarrow$ graph repair.

Morph operator	Triggered by	Modifies	Mechanism
pruning.morph()	Gate 1 REJECT	G_R, G_S	Deactivates EventNodes with $N_i \geq 0.3$; deactivates AgentNodes with no event coverage. Lowers N_{global} below Gate 1 threshold.
verification.morph()	Gate 2 BLOCK	G_S, G_R	Inserts a VERIFIER AgentNode into G_S ; adds projection child EventNodes in G_R for high- σ nodes. Raises coherence c above c_{min} .
ensemble.morph()	Gate 3 RE_ENCODE	G_S, G_R	Adds k solver variants around the deficient AgentNode plus an aggregator; rewires G_S edges. Adds normalization child EventNodes for high- σ events in G_R . Raises κ_{gate} and lowers σ .
event.expansion.morph()	Gate 4 REPAIR	G_R only	Adds higher-quality modal EventNodes (higher κ_L) as children of ungrounded nodes. Checks projected N_{dst} before insertion to avoid triggering Gate 1. Raises κ_L above $\kappa_L^{\text{min}} = 0.3$.

Remark B.1 (Convergence Guarantee). *Each morph operator is designed to move exactly one certificate*

scalar in one direction: pruning_morph lowers N ; verification_morph raises c ; ensemble_morph raises κ_{gate} and lowers σ jointly; event_expansion_morph raises κ_L . No morph degrades a certificate that was already passing. This property ensures that the adaptive execution loop `run_dual_graph` terminates: if a certificate deficit is structurally repairable, at most one morph per deficient axis is required. If it is not repairable (e.g., the surjectivity starvation case where `pruning_morph` is a no-op), the loop detects this condition and halts with a logged reason rather than cycling.

B.5 Surjectivity and the Curriculum Constraint

The φ -morphism must be surjective: every `AgentNode` in G_S must be reachable by at least one `EventNode` in G_R . An agent with no event coverage has no grounding — it is a solver whose compatible encoding simply does not exist in the current representation graph. DGM enforces surjectivity by assigning such agents a forced local certificate ($N = 0.8, \sigma = 1.0$), which immediately triggers Gate 1 REJECT, which triggers `pruning_morph`.

However, if no `EventNode` has $N_i \geq 0.3$ at the time of pruning (because the noise source is structural, not content-level), `pruning_morph` removes no nodes and the surjectivity deficit persists. This is the *surjectivity starvation* failure mode: the loop fires Gate 1, calls `pruning_morph`, finds nothing to prune, re-derives the same certificate, and repeats. Detection:

```

1: if pruning_morph is called then
2:    $n_{\text{before}} \leftarrow |V_R^{\text{active}}|$ 
3:   Execute pruning_morph()
4:    $n_{\text{after}} \leftarrow |V_R^{\text{active}}|$ 
5:   if  $n_{\text{after}} = n_{\text{before}}$  then
6:     Log: SURJECTIVITY STARVATION — no pruneable events
7:     break ▷ Halt repair loop; deficiency is structural
8:   end if
9: end if
```

The curriculum alignment check is a joint constraint on (G_R, G_S) : both graphs must belong to the same curriculum phase Φ . Phase 3 G_R nodes (high σ , rich evidence DAG) wired to a Phase 1 G_S (narrow capability centroids, linear pipeline) produce the *impedance mismatch* failure mode: σ is elevated by construction, Gate 3 fires, `ensemble_morph` adds Phase 1 variants, σ remains elevated because Phase 1 centroids cannot span the Phase 3 event space, and the loop repeats. The circuit breaker is `validate_phi_curriculum_alignment()`, which detects the mismatch before the execution loop starts.

Remark B.2 (Joint (G_R, G_S) Construction Requirement). *The curriculum phase is a constraint on the joint valid region of (G_R, G_S, φ) space. Adjusting only G_S (agent `SolverContracts`) while leaving G_R at synthetic defaults produces a φ -morphism that is tight by construction but meaningless, because neither graph represents actual phase-appropriate difficulty. Both graphs must be constructed for the same phase. This is not an implementation detail; it follows from the RSCT requirement that the triple $(\alpha, \kappa_{\text{gate}}, \sigma)$ be computed from the actual encoding policy $\mathcal{M}$ applied to actual problem instances, not from proxy defaults.*

B.6 The Two-Direction Framework Integration Problem

DGM can be wired to real agent frameworks (CrewAI, A2A, Swarm-It) but doing so correctly requires understanding that the wiring is bidirectional. Both directions are necessary; neither alone is sufficient.

Definition B.4 (Direction 1 — Input: Real Content $\rightarrow G_R$). *Real framework content (user queries, task outputs, retrieved evidence, inter-agent messages) must be translated into EventNode objects with real YRSN certificates before the gatekeeper runs. This requires an encoder that reads actual content and computes $(R_i, S_{\text{sup}_i}, N_i)$ from it, not from hardcoded phase defaults.*

If Direction 1 is absent, the gatekeeper certifies phantom data. When a downstream agent returns a hallucinated, contradictory output, N_i is not updated, Gate 1 never fires on it, and the next agent is certified against the same synthetic events as the first. The gatekeeper is noise, not signal.

Definition B.5 (Direction 2 — Output: EXECUTE $\rightarrow$ Framework Fires). *When the gatekeeper returns EXECUTE, the `impl` field of the certified AgentNode fires the underlying framework. The `impl` function is the only mechanism by which a DGM EXECUTE decision causes a real LLM call, A2A routing decision, or Swarm-It worker invocation.*

Remark B.3 (Why Direction 2 Alone Is Insufficient). *Wiring only Direction 2 (an `impl` function that fires the framework on EXECUTE) gives the gatekeeper control over when the framework runs but not over what it certifies. The gatekeeper still certifies the same synthetic G_R events across all invocations. Each agent in a multi-agent pipeline is certified against the original events rather than the actual intermediate content produced by its predecessors. The EXECUTE chain runs — it just runs on phantom certificates throughout.*

Direction 2 without Direction 1 is equivalent to a security checkpoint that waves every car through based on the first car’s registration, regardless of what changed.

The correct implementation is the `RSCTAgentNode`, which integrates both directions in a single `impl` wrapper:

1. On EXECUTE: fire the underlying framework (Direction 2).
2. Receive the real framework output.
3. Translate the real output into a new EventNode via the content encoder (Direction 1: real content $\rightarrow G_R$).
4. Add the new EventNode to G_R .
5. The next agent’s `derive_local_certificate()` call will read this real EventNode and return a real certificate.

This ensures that each gatekeeper invocation in a pipeline certifies the actual intermediate state of the system, not the state at initialization.

B.7 YRSN Estimation Fidelity

Direction 1 requires a content encoder that produces (R_i, S_i, N_i) from actual text, image, or structured content. Three fidelity levels are available, in increasing order of accuracy:

Table 6: YRSN estimation fidelity levels for real framework content.

Level	Dependency	Method
1 (Heuristic)	None	Keyword density proxies: noise signal words (“error”, “contradicts”, “uncertain”) contribute to N_i ; hedging words (“possibly”, “might”) contribute to S_{sup_i} ; length variance relative to inputs proxies σ_{local} . Usable immediately; low accuracy.
2 (Semantic)	sentence transformers	Cosine similarity between content embedding and query embedding estimates R_i directly. Unlocks meaningful σ and κ_{int} values. Required before σ -dependent theorems (H2, H3, Oobleck) produce meaningful experimental results.
3 (Full YRSN)	yrsn core certificates	Direct call to the certified YRSN encoder. Produces exact $(R_i, S_{\text{sup}_i}, N_i)$ from arbitrary content under the simplex guarantee $R + S_{\text{sup}} + N = 1$. Production target.

Remark B.4 (Why Level 1 Invalidates σ -Dependent Experiments). *A hash-based encoder produces $\sigma \approx 0.008$ for all inputs regardless of content, phase, or modality. This is not a numerical coincidence; it is a structural consequence of hash-based embeddings having no semantic geometry. The $\lambda\sigma$ term in the Oobleck threshold contributes ≈ 0.003 to κ_{req} , making Gate 3 effectively static. Any experiment designed to test σ -variation (H2, H3, Oobleck accuracy, morphing improvement rate) will produce random-chance results until Level 2 or Level 3 estimation is in place. Threshold recalibration to compensate for a flat σ is not a valid workaround; it papers over the missing semantic signal.*

B.8 Audit Trail and SR 11-7 Compliance

Every gatekeeper invocation in DGM emits a structured audit log entry:

$$\mathcal{L} = (\alpha, \omega, \alpha_\omega, \kappa_{\text{gate}}, \sigma, \text{gate_reached}, \text{outcome}, t_{\text{stamp}}). \quad (22)$$

This tuple directly satisfies SR 11-7 §4(c) requirements: α and ω document data quality and distributional coverage; κ_{gate} and σ satisfy sensitivity analysis and stability monitoring; gate_reached and outcome provide timestamped documentation of model override decisions. Aggregated over time, the empirical distribution of $(\kappa_{\text{gate}}, \sigma)$ constitutes the back-testing record required by SR 11-7 §4(c) — a quantitative basis for model recalibration triggers that does not rely on post-hoc output review.

The audit trail must be written *before* any non-EXECUTE halt, not after. If an exception is raised before the audit entry is committed, the halt is not traceable and the SR 11-7 record is incomplete. This is why `CurriculumAwareDGM` returns an empty `ExecutionContext` with a committed audit entry on curriculum mismatch rather than raising a bare exception.